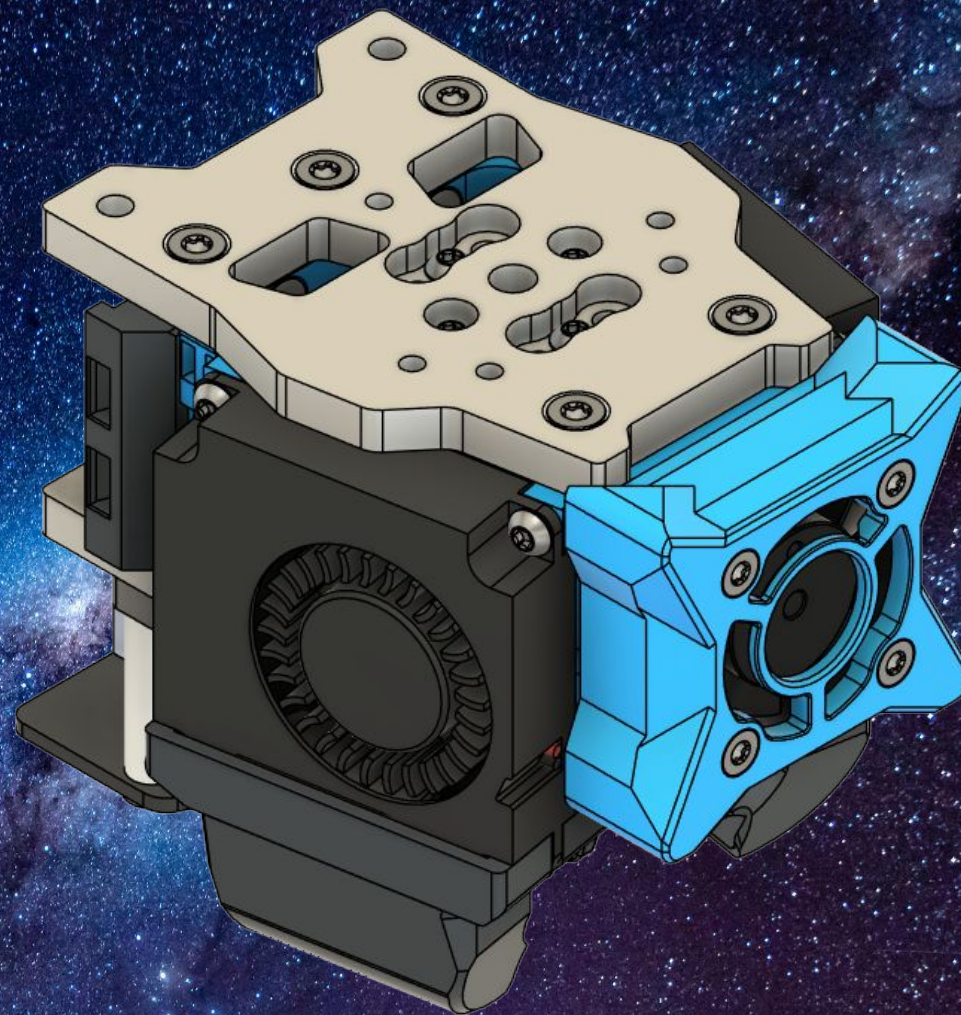




HYPERNOVA BUILD GUIDE

"We build precision with simplicity, so anyone can create the impossible."

VERSION 2025-09-09

Before beginning your build, please take a few minutes to carefully check all of the included hardware. Verifying your kit ensures a smooth assembly process and prevents delays caused by missing parts.

Included Hardware:

- M3x40 Countersunk Torx — 5 pcs
- M3x10 Countersunk Torx — 3 pcs
- M3x20 Button Head Torx — 4 pcs
- M3x12 Countersunk Torx — 4 pcs
- M3x8 Button Head Torx — 5 pcs
- M3x26mm Standoff — 2 pcs
- M3x8 Countersunk Torx — 4 pcs
- M3x8 Ultra Low Profile Torx — 2 pcs
- M2.5x6 Button Head Torx — 8 pcs
- M2x10 Socket Head — 2 pcs
- M2.5x16 Countersunk Torx — 4 pcs
- M2.5 Heat-sets — 10 pcs
- M3 Heat-sets — 13 pcs
- M3 Hyper Sets — 3 pcs
- 4010 Fans — 2 pcs
- 2510 Hyper Fan — 1 pcs
- Top Plate HN-TTPU — 1 pcs
- Bottom Plate HN-BPU — 1 pcs

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HYPERNOVA PRINT GUIDE

Use the following print settings in your slicer of choice

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

MATERIAL

ABS/ASA

LAYER HEIGHT

Recommended: 0.2mm

EXTRUSION WIDTH

Recommended: Forced 0.4mm

INFILL TYPE

Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE

Recommended: 50%

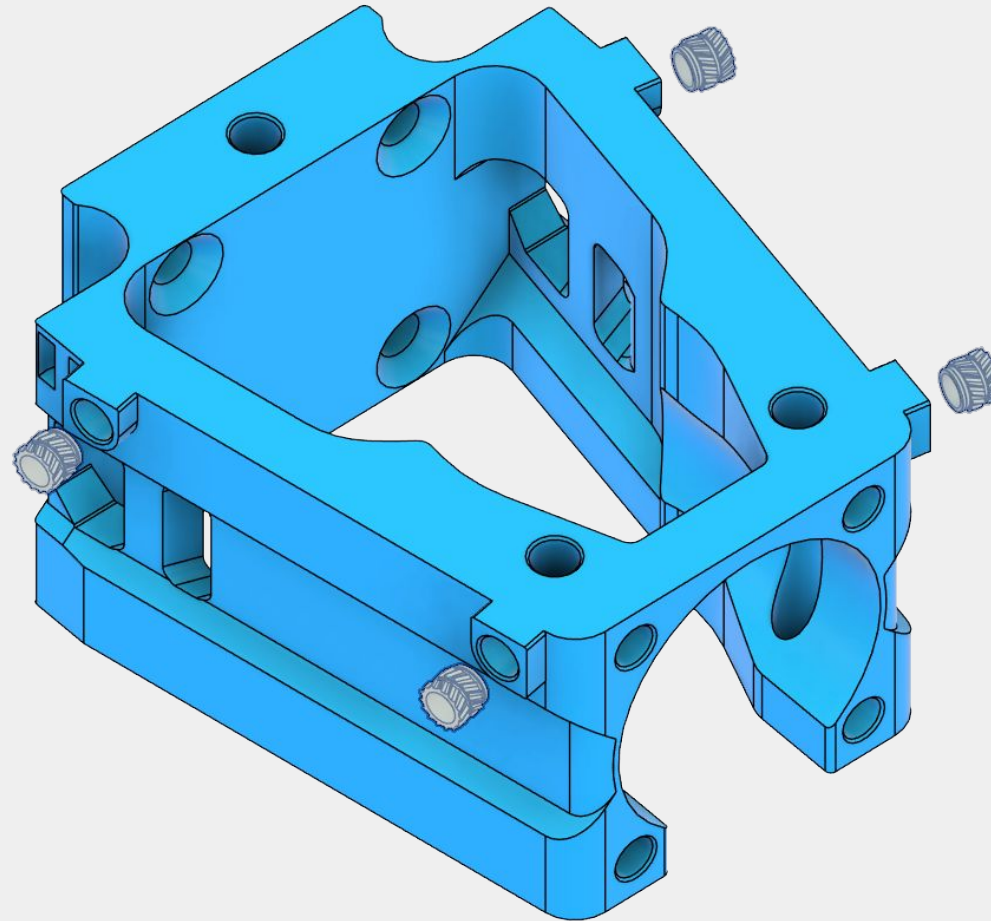
WALL COUNT

Recommended: 4

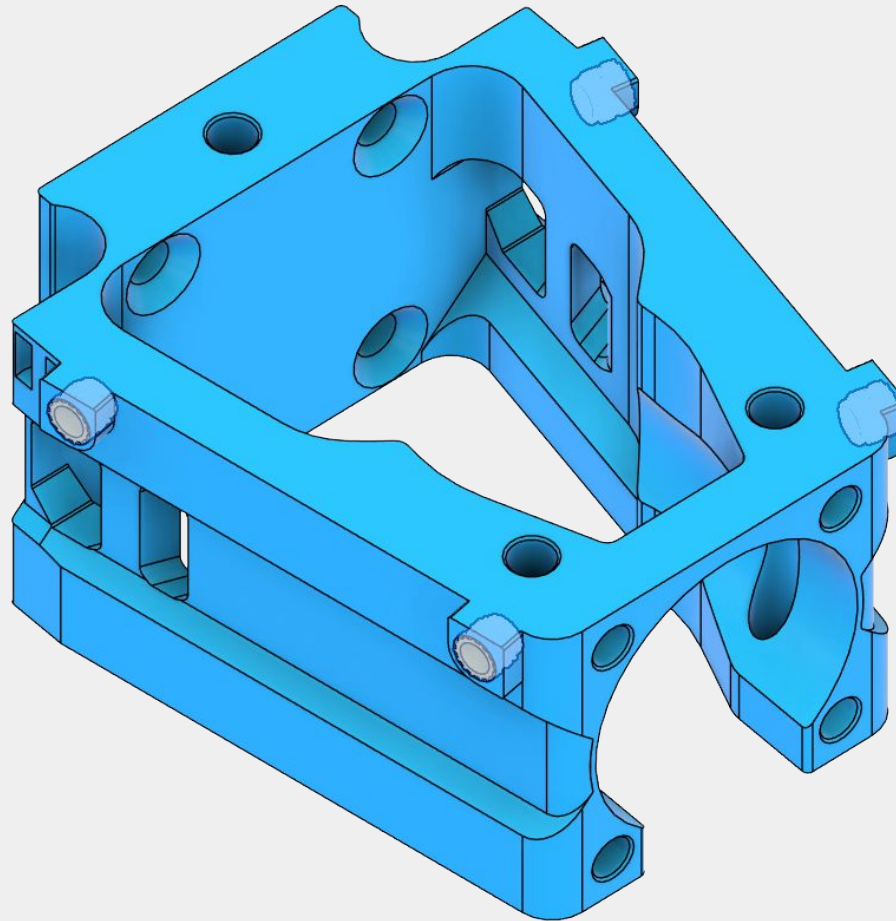
SOLID TOP/BOTTOM LAYERS

Recommended: 5

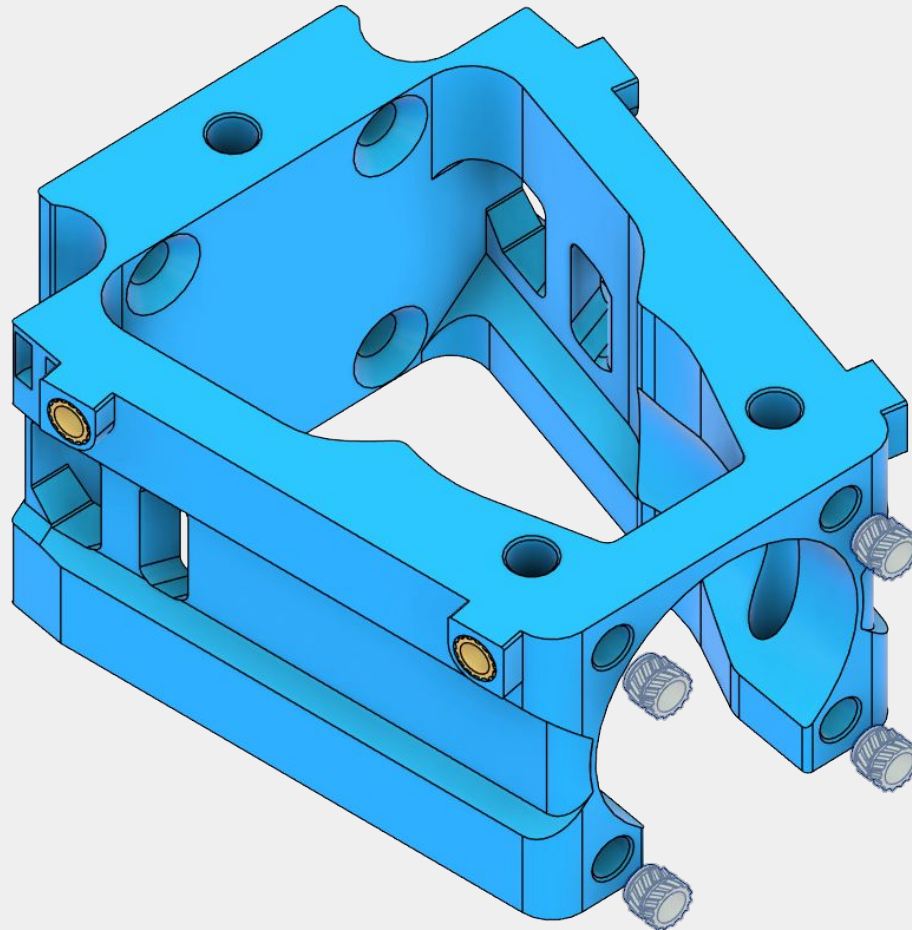
Melt in all four M2.5 heat sets in the side core, use 250°C to not over melt the plastic.



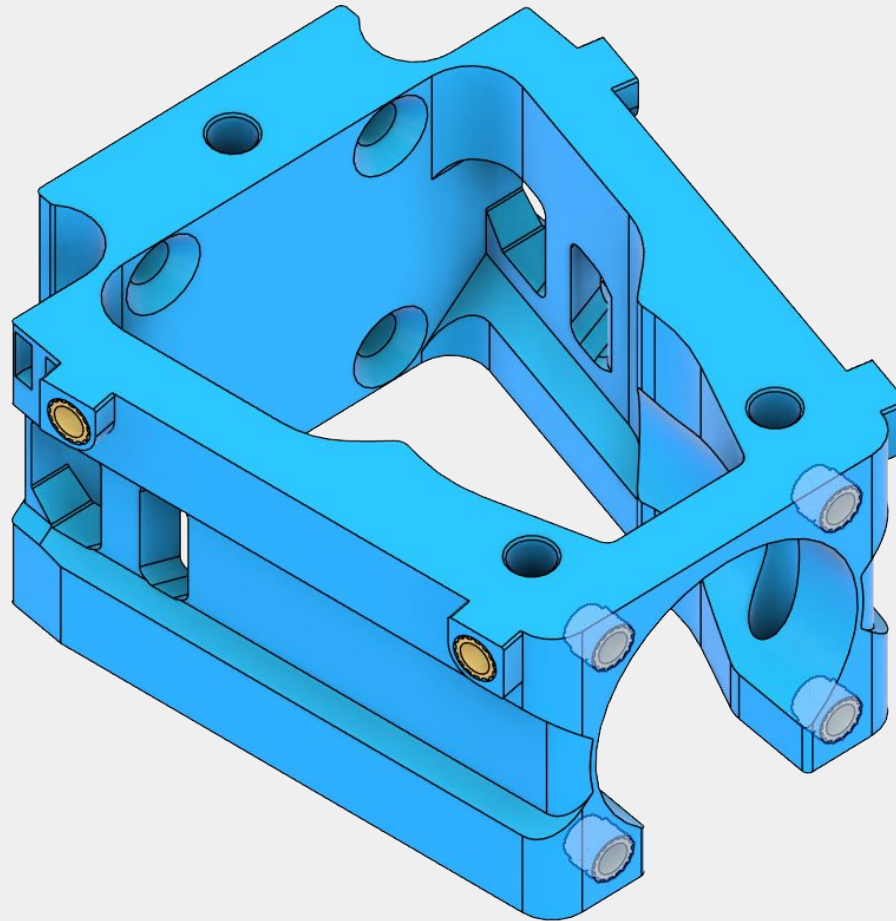
Make sure the heat sets are flush and install not at an angle.



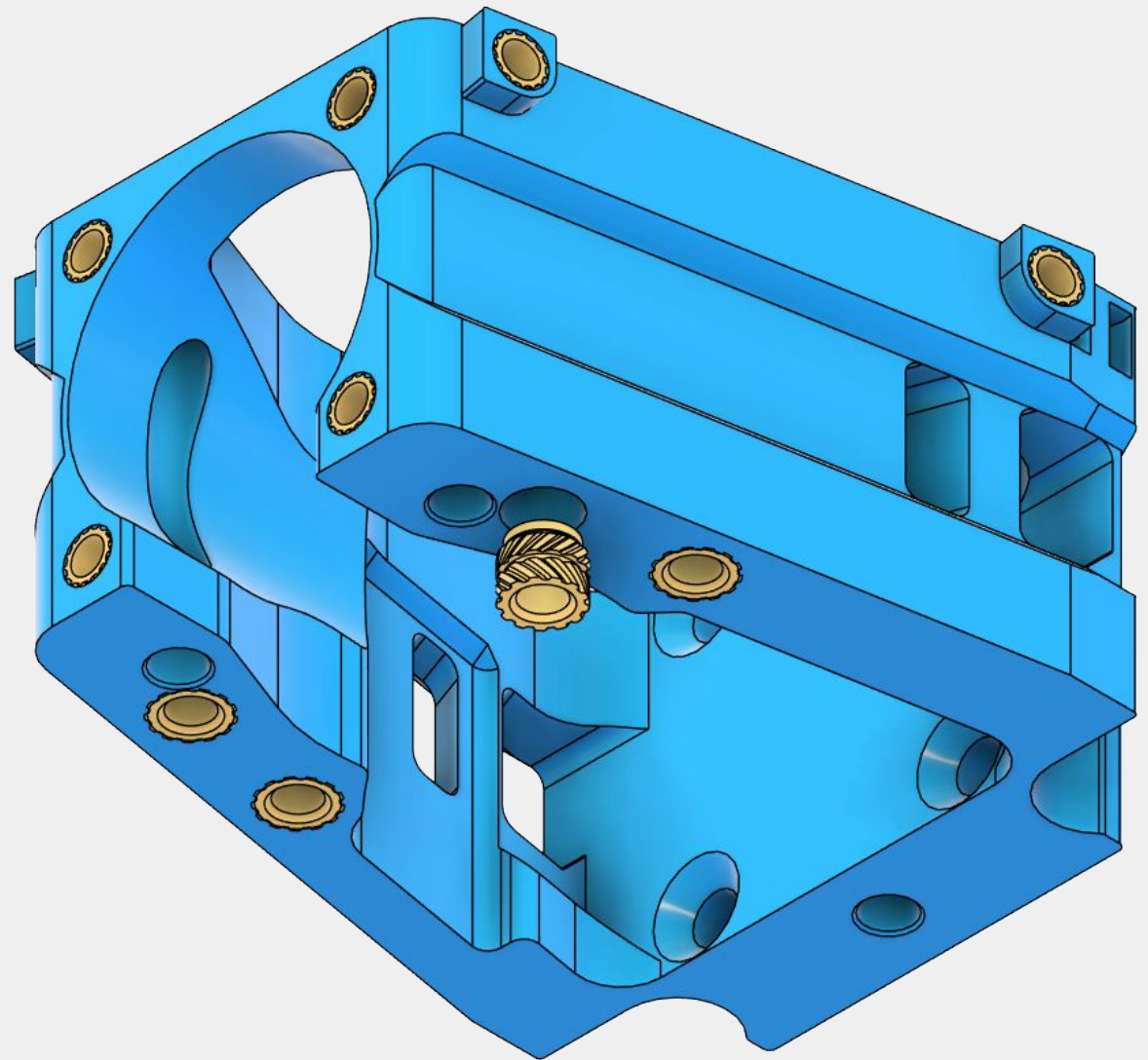
Melt in all four M2.5 heat sets in the front of the core, use 250 C to not over melt the plastic.

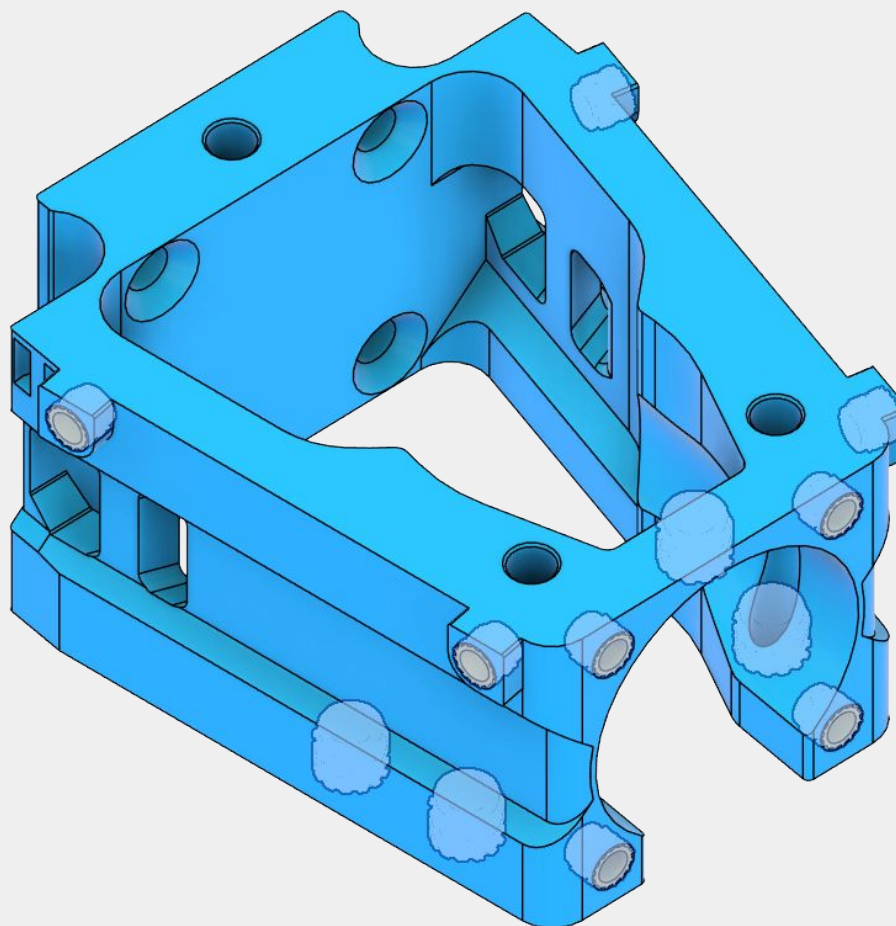


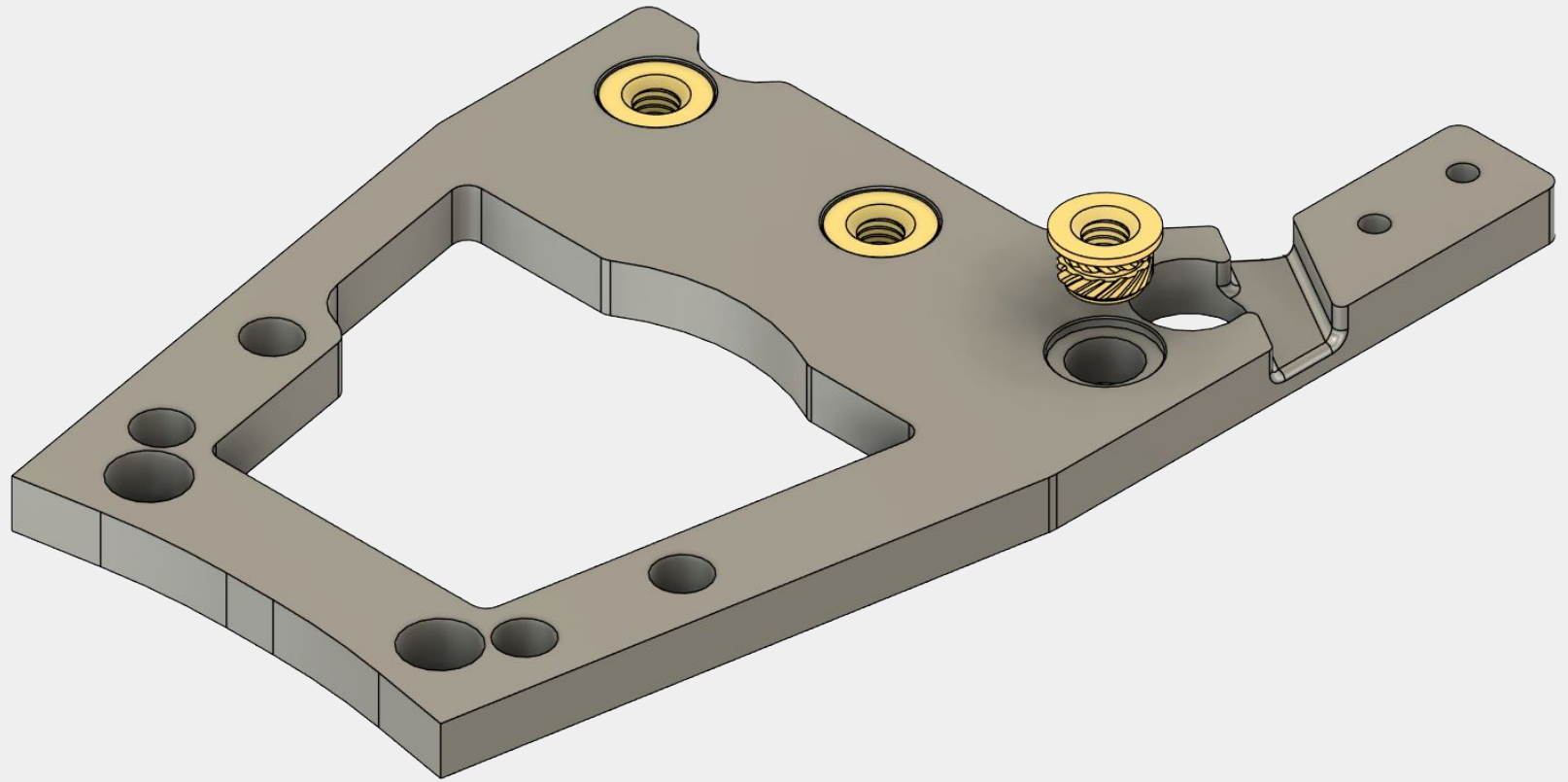
Make sure the heat sets are flush and install not at an angle.



Install four M3 heat-sets into the core.

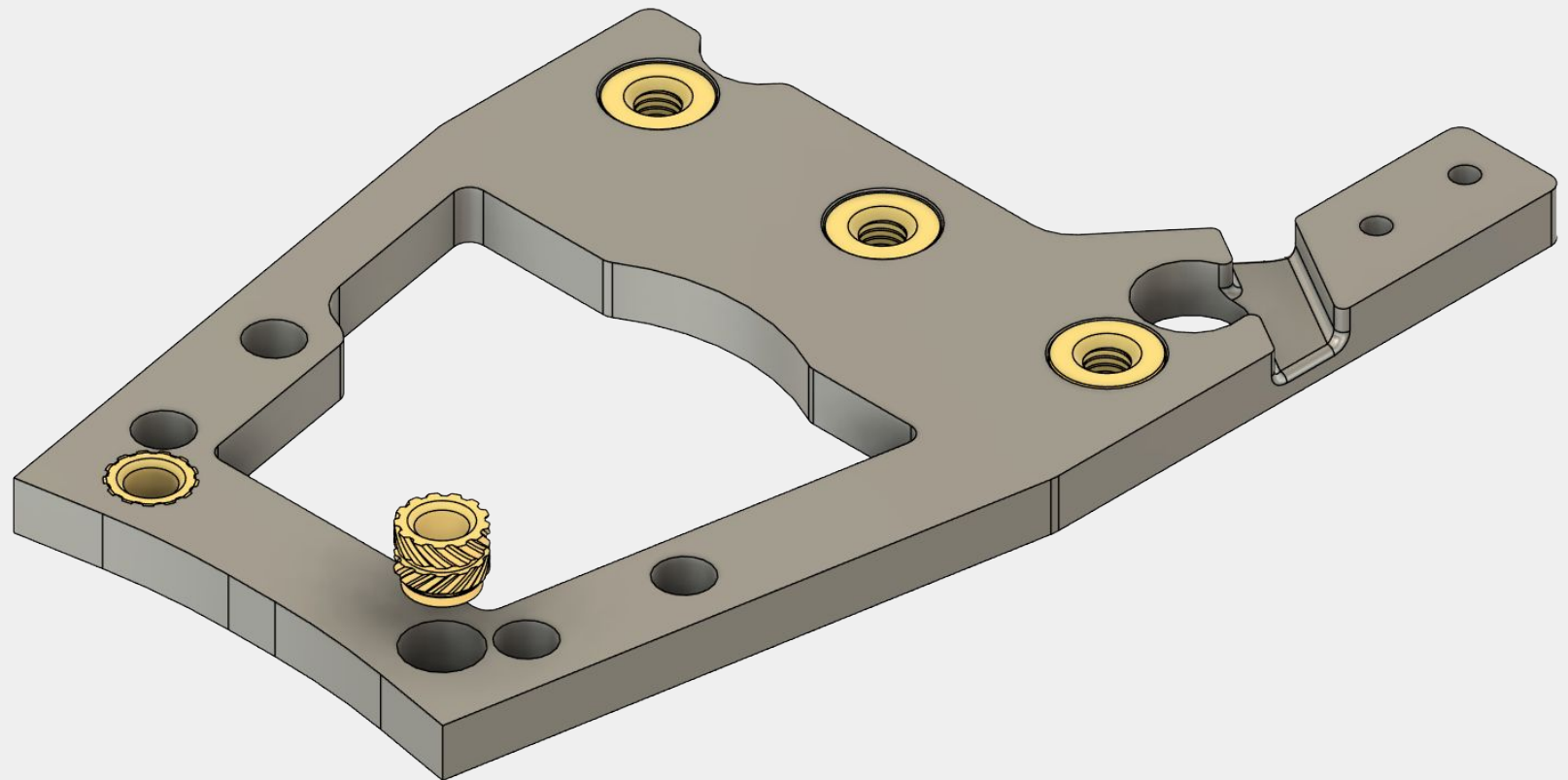




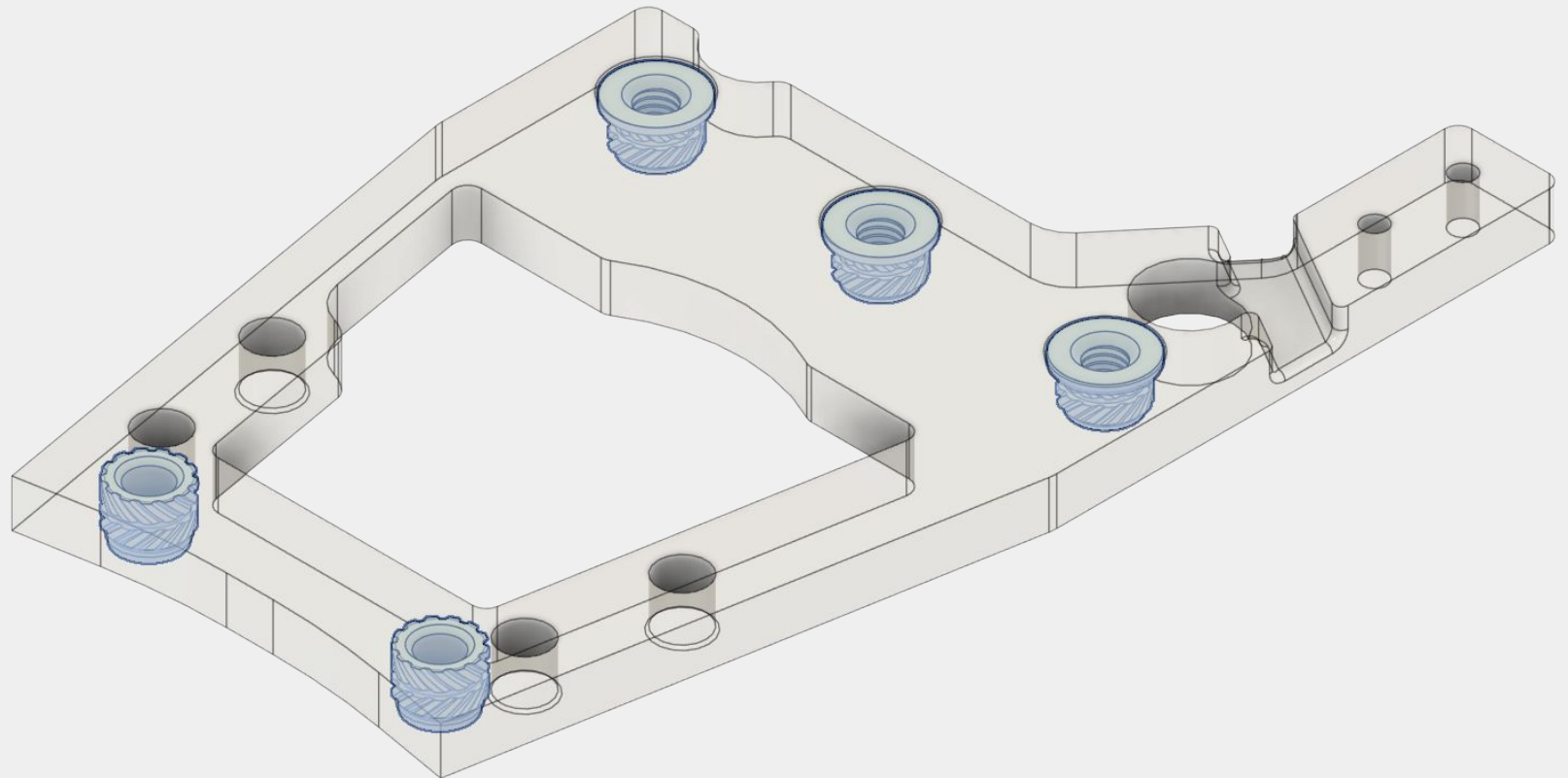


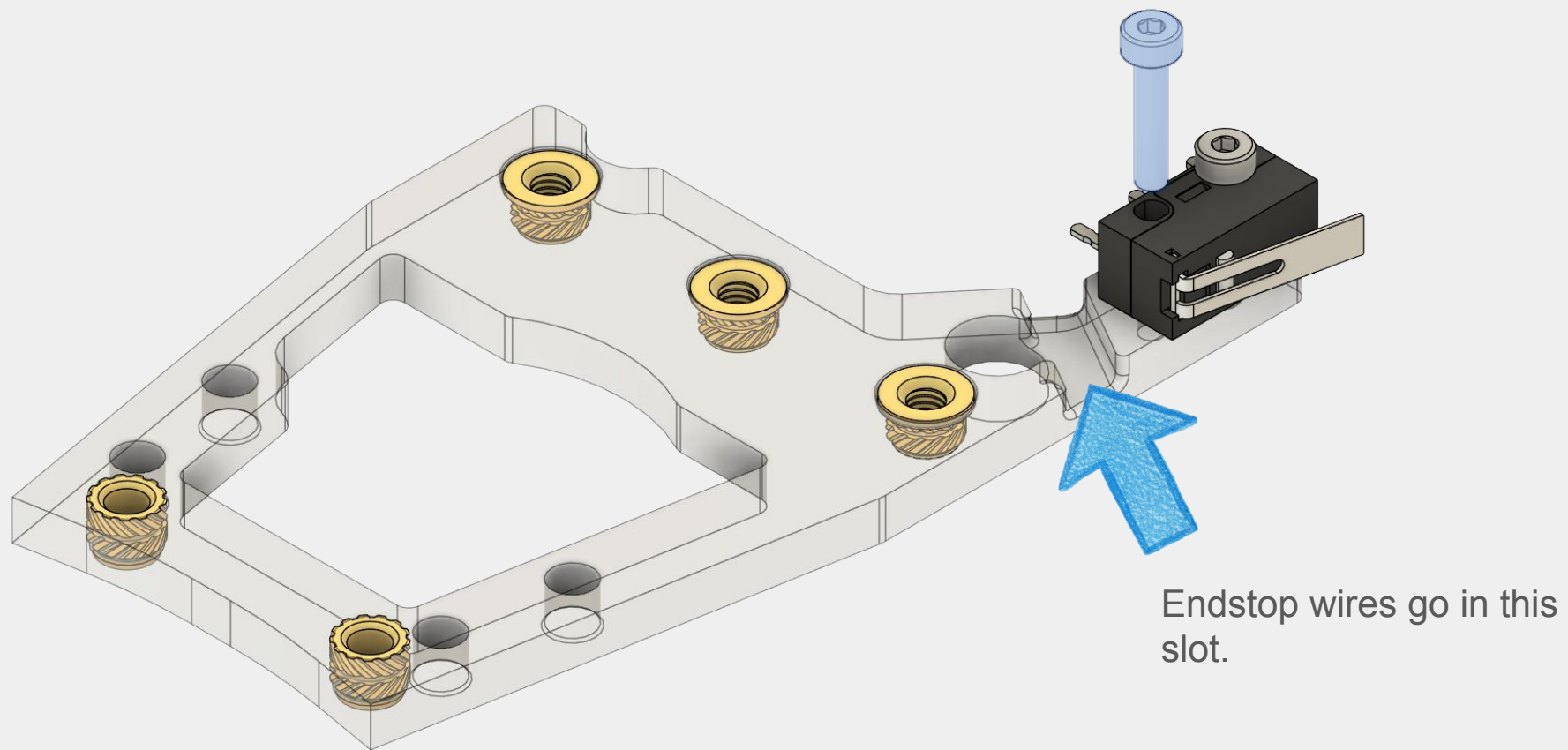
Note: HYPERSETS™ are a flanged form of a heat set insert.

Three-HYPERSETS

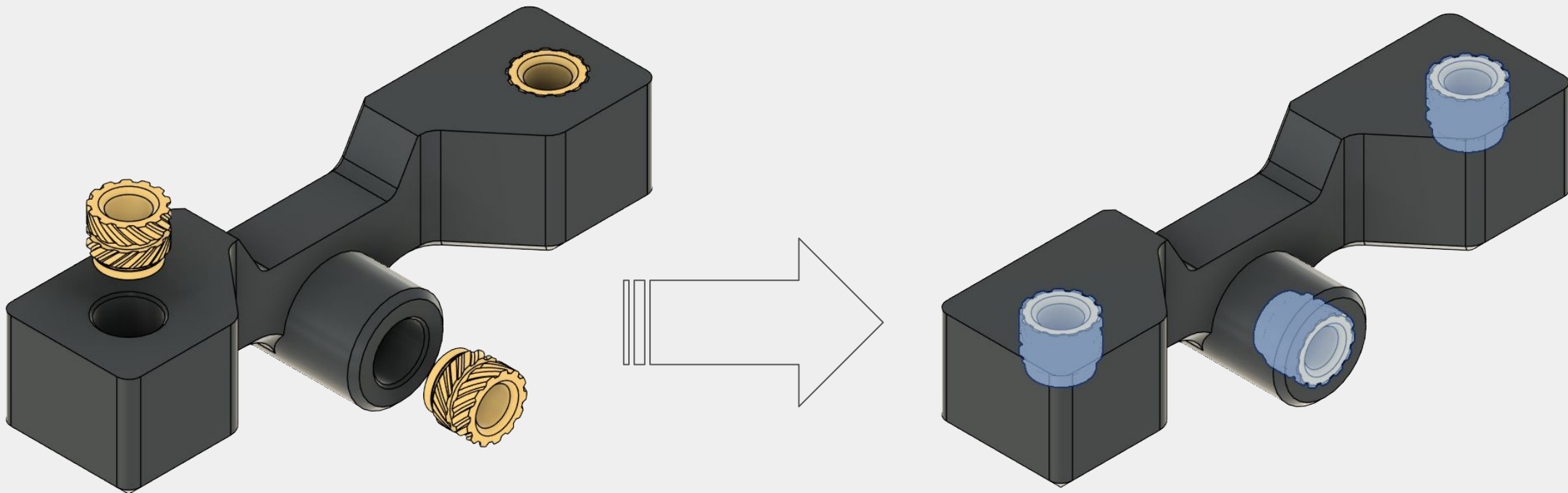


2-M3 HEATSETS

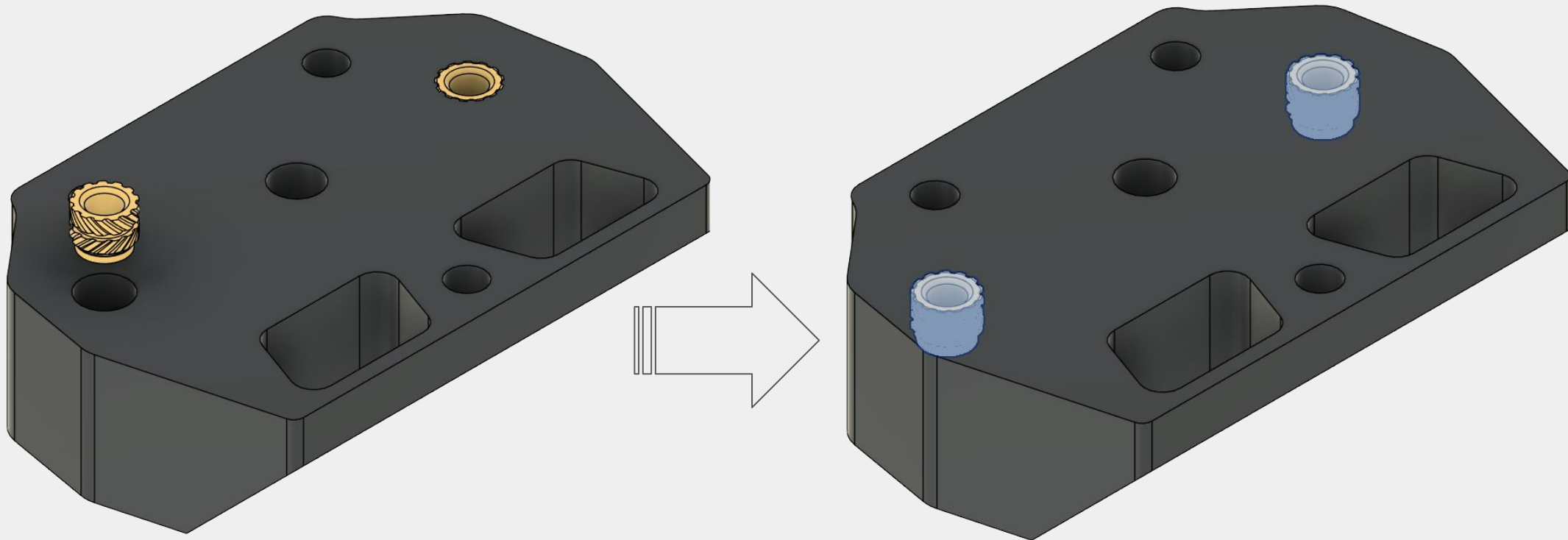




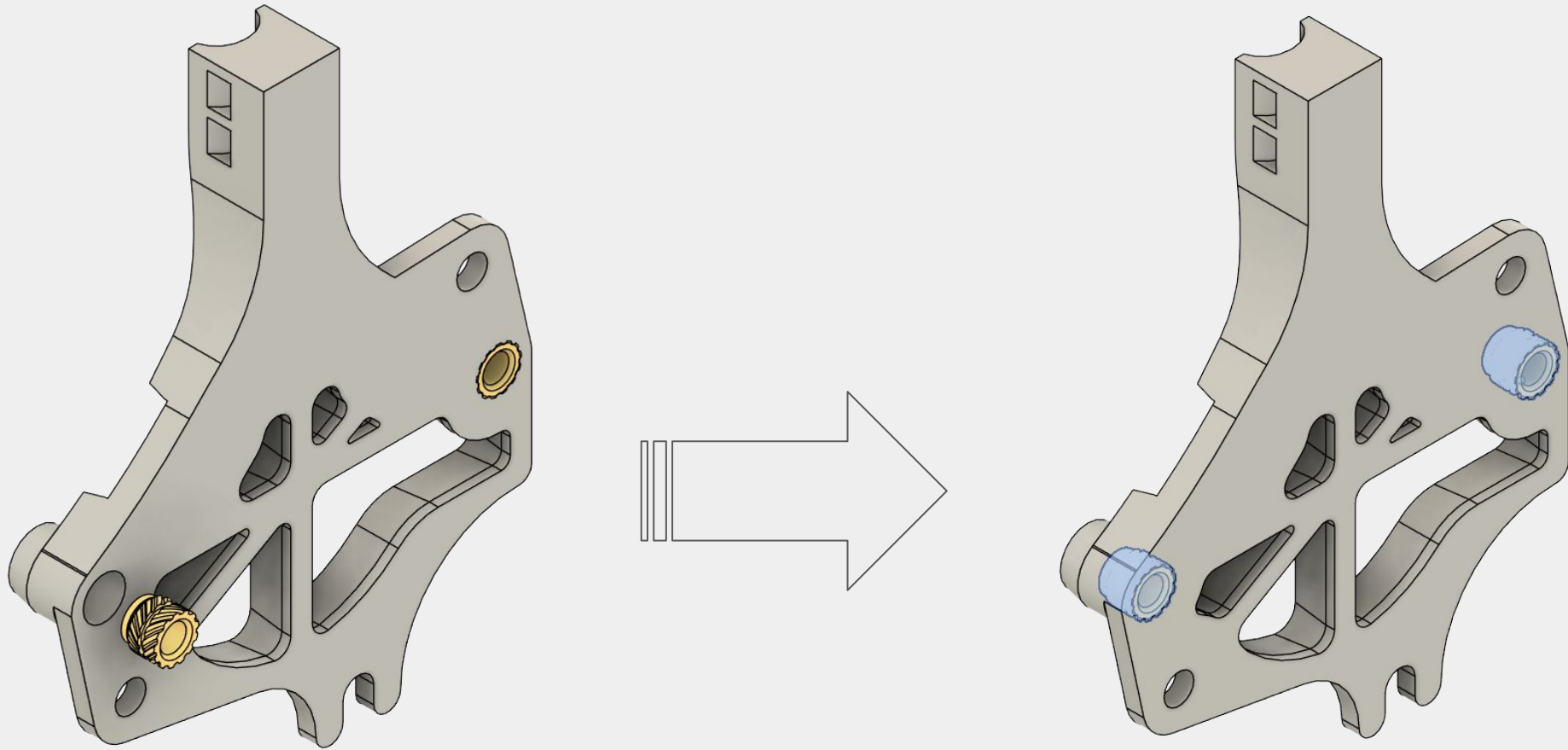
2-M2X10MM SHCS



Three-M3 heat-sets



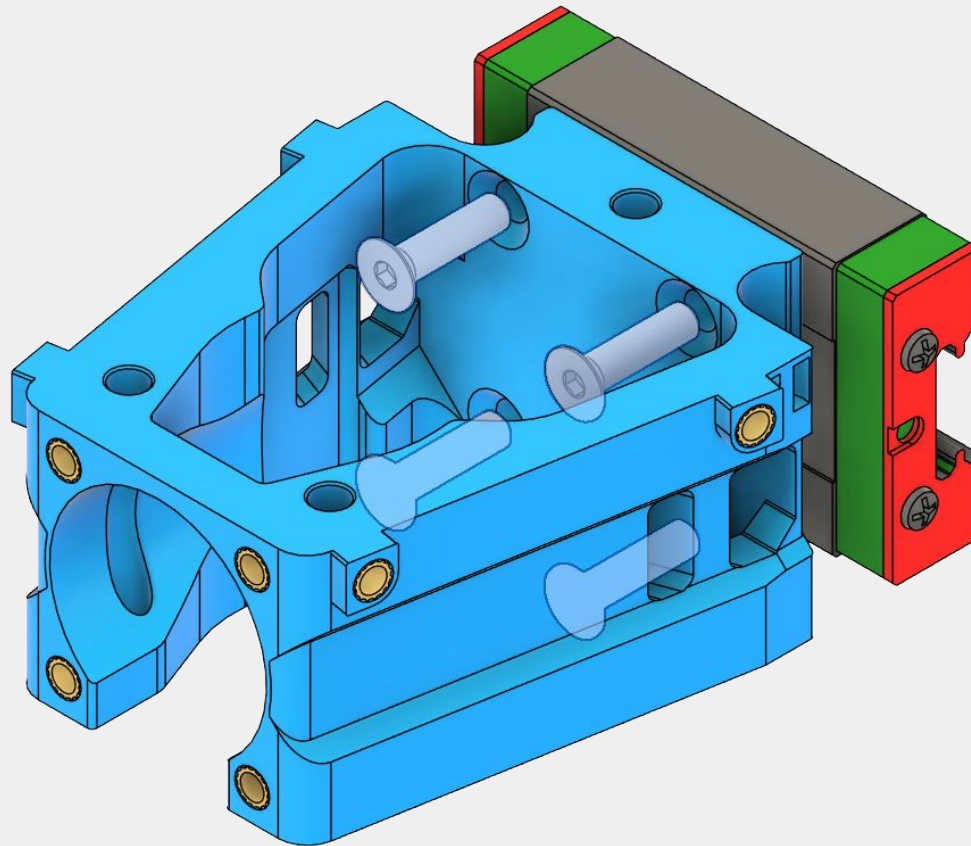
Two-M3 HEAT-SETS



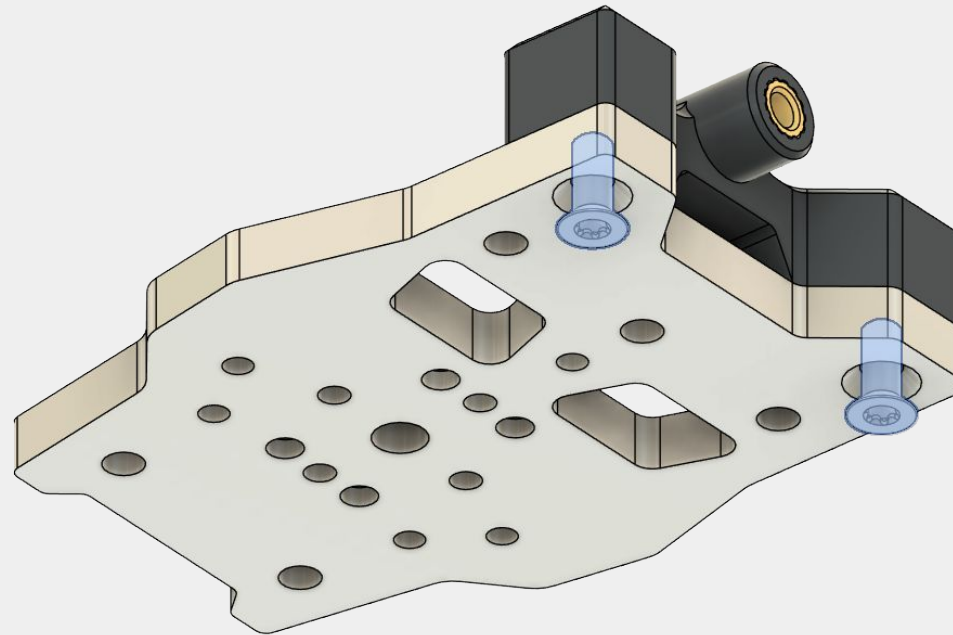
Note: Sherpa mini can mount does not need heat sets.

Two-M3 HEAT-SETS

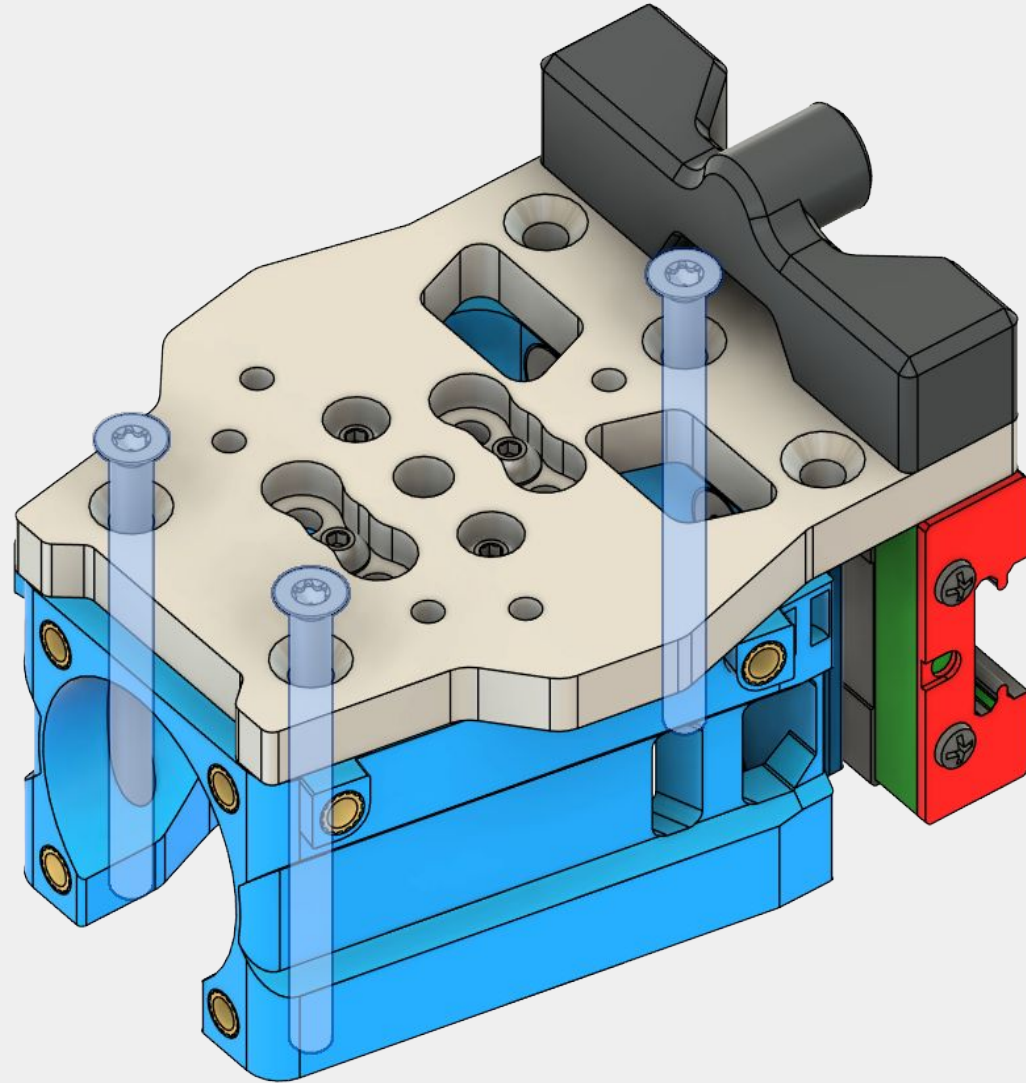
Mount core with
four M3x12 screws.



Mount canbus brace
with two M3x8
countersunk screws.

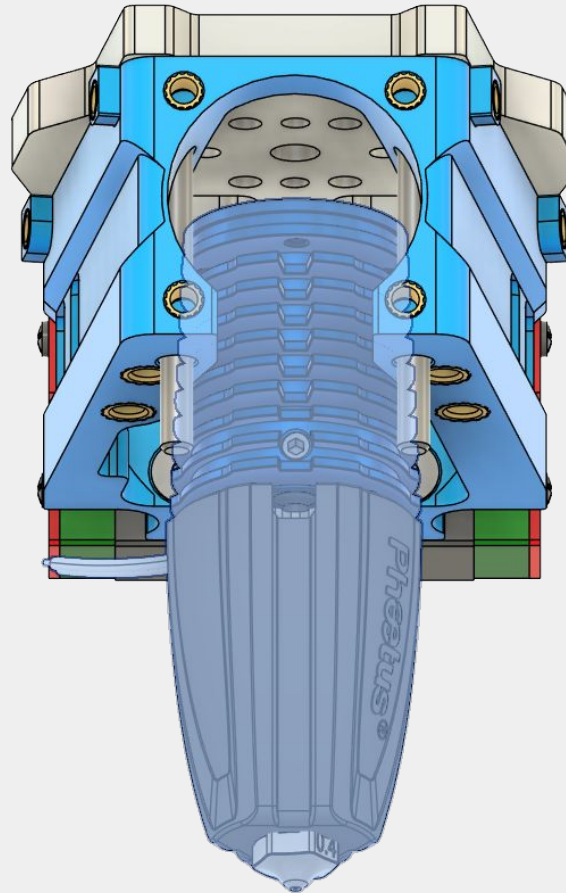


Insert three of the five
M3X40mm countersunk
Screw through the top
plate.

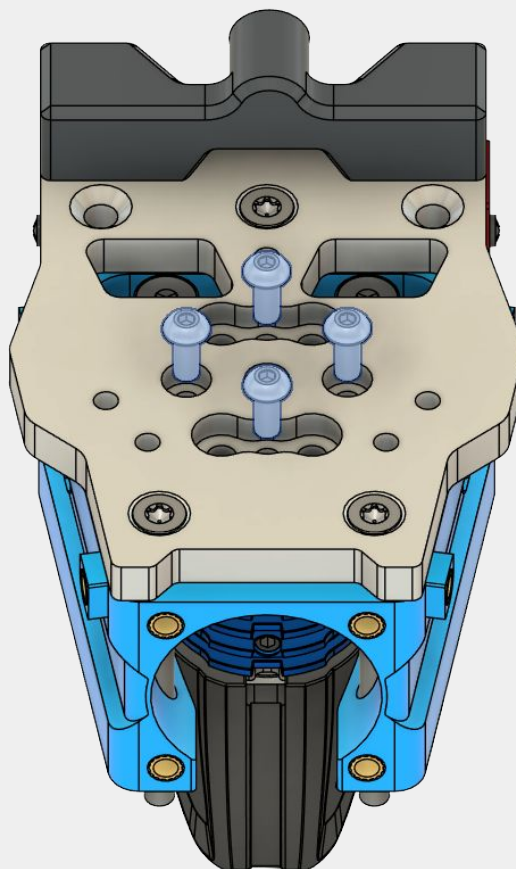


Align hotend into the core with wire exiting on the left.

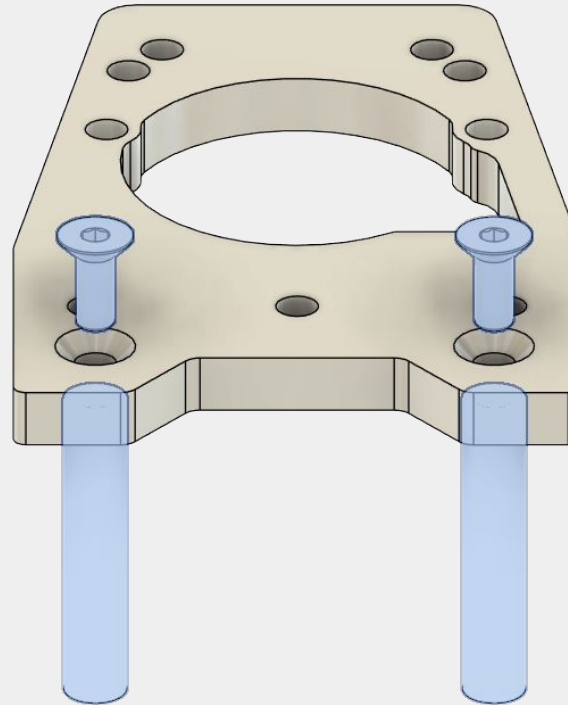
Note: you may need to rotate the Heat Block for the wire to exit to the correct position.



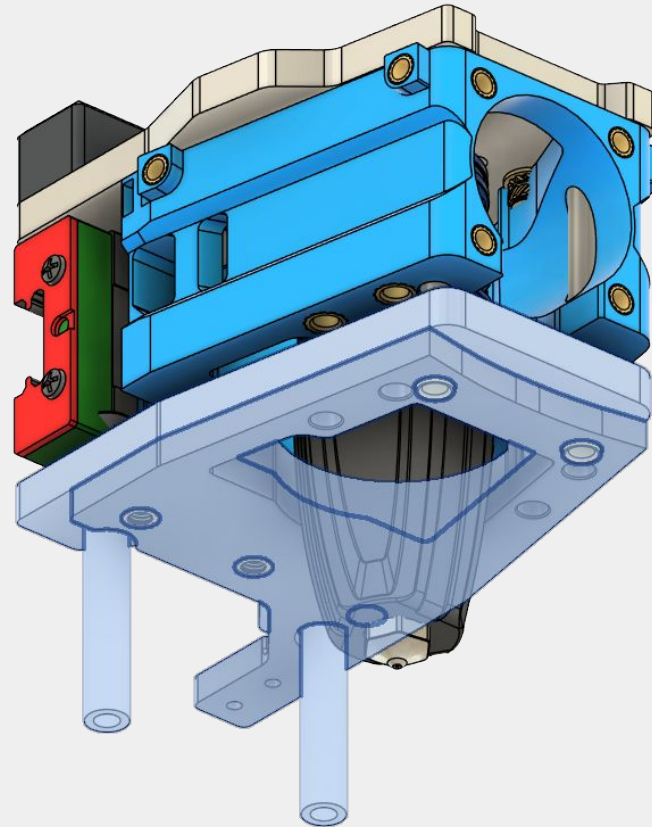
Mount Hotend with
4 M2.5x6 screws.

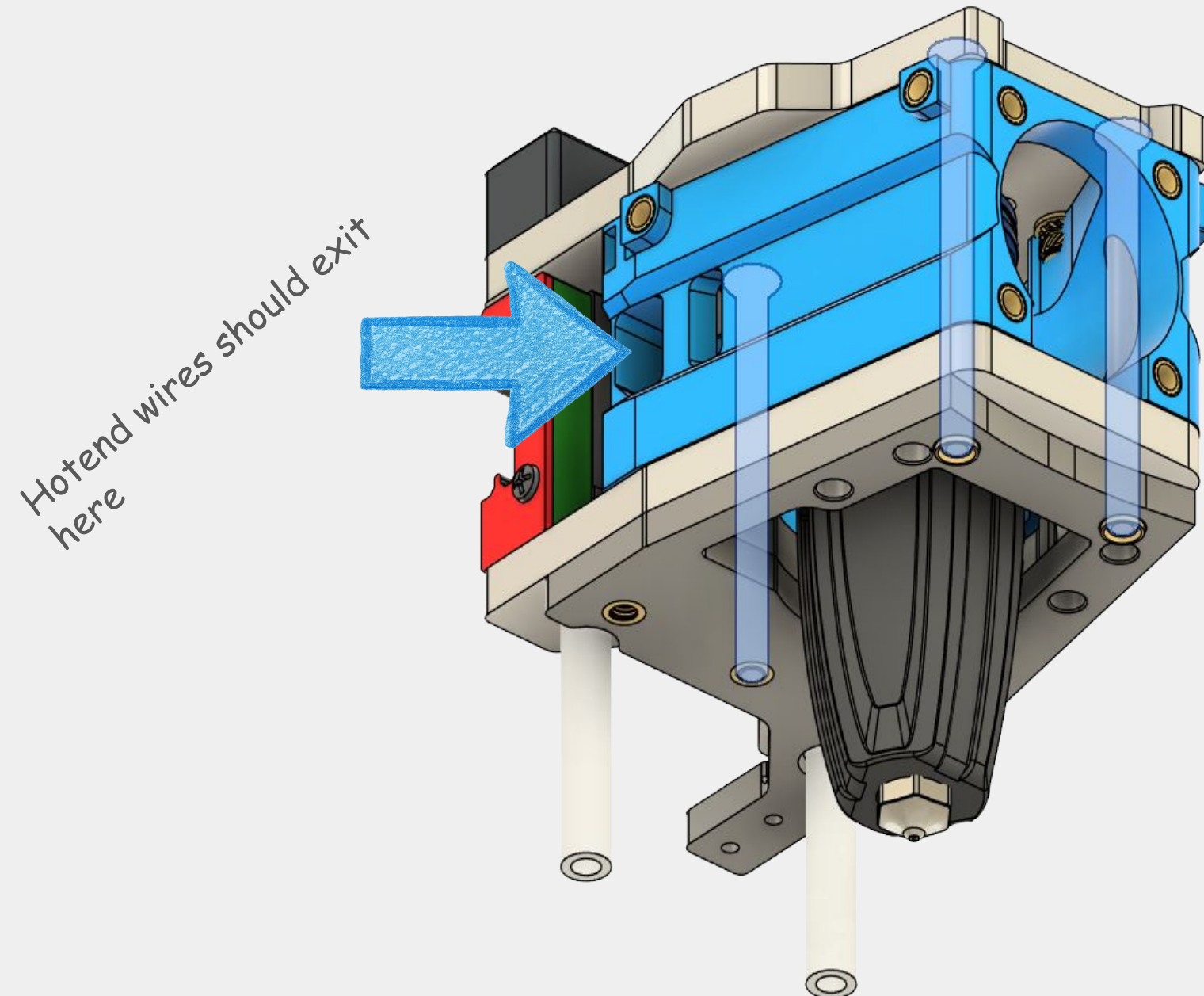


Install two M3X8 countersunk torx into the beacon standoffs.



Position the bottom plate and spacer to the core.

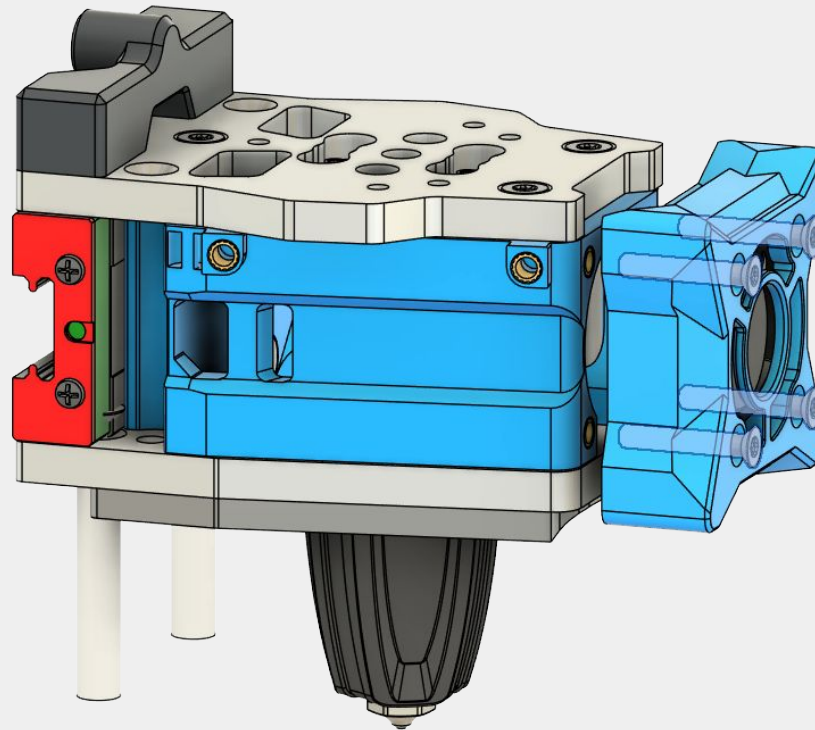




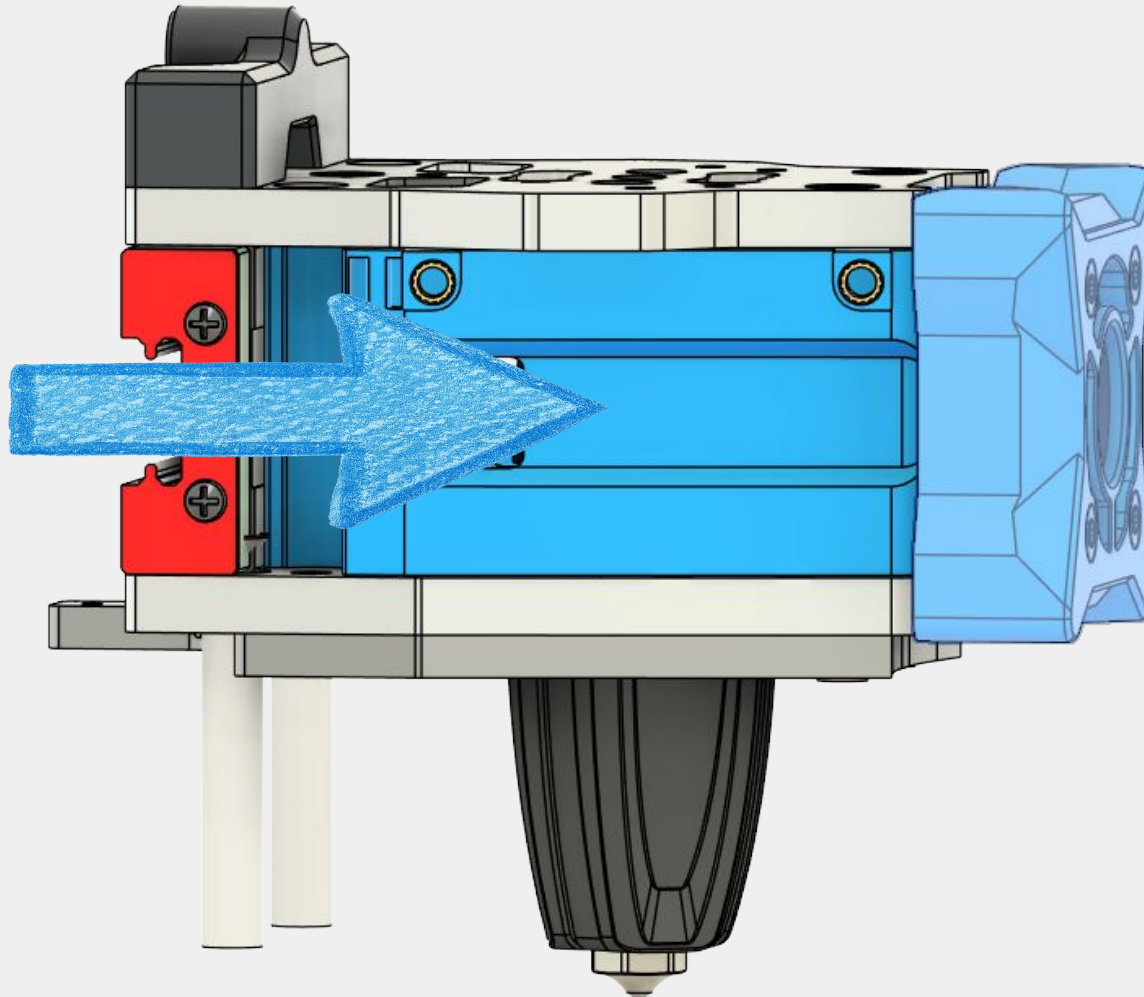
INSTALL 2510 FAN AND COVER

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Install the 2510 fan in
the cover then install
four M2.5x16
Countersunk Torx.



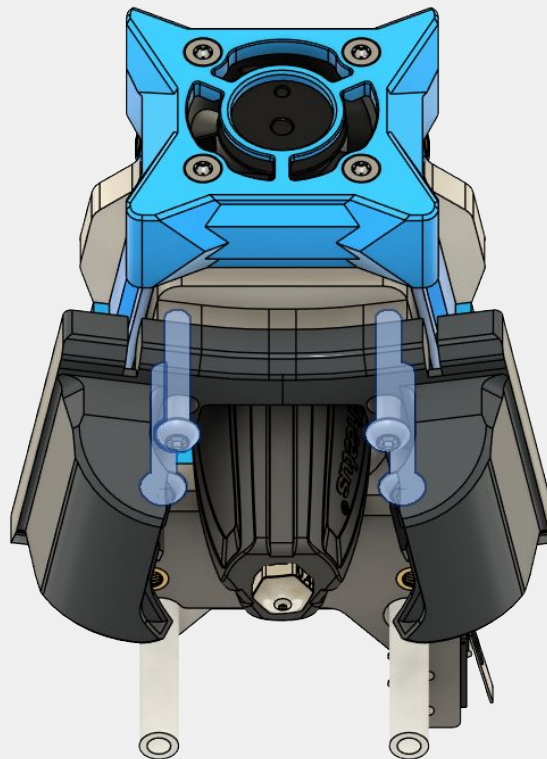
USE FOR FAN
WIRES



MOUNT FAN DUCTS

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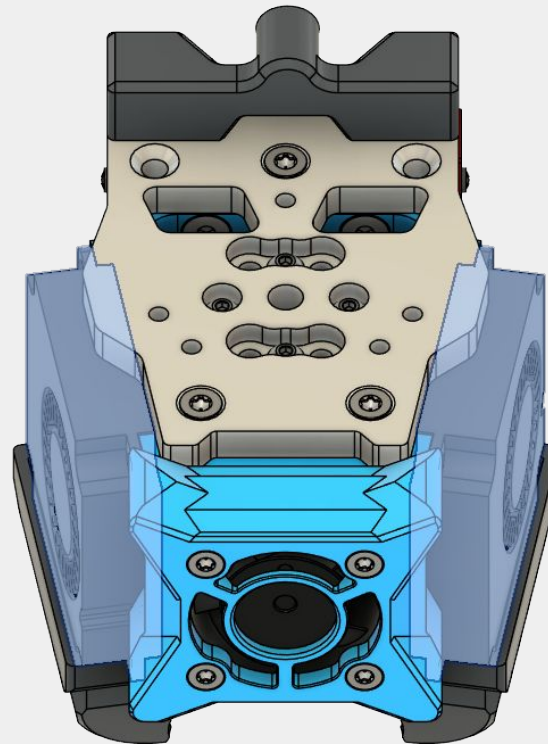
Install four M3x20
button head Torx.



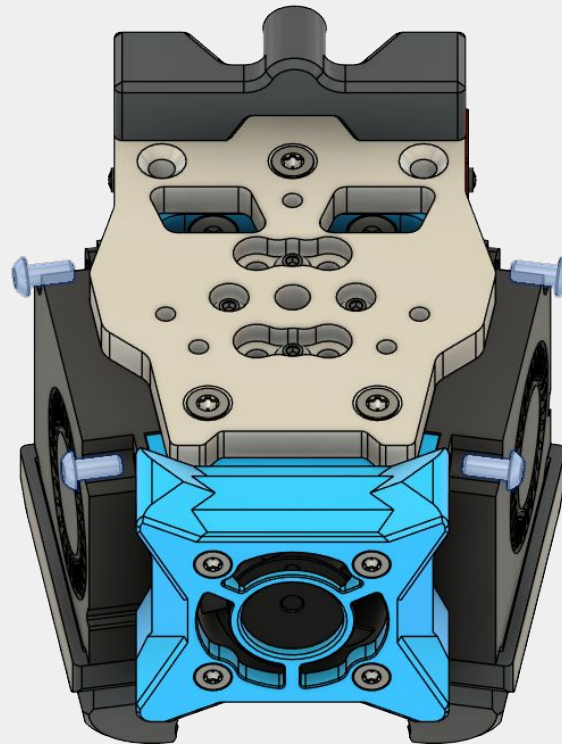
4010 FAN MOUNTING

Angle the 4010 fans into the ducts first and press the fans into the core.

Note: you may need to loosen the fan duct screws for the fan to fit.



Install four M2.5x6
Button Head Torx.

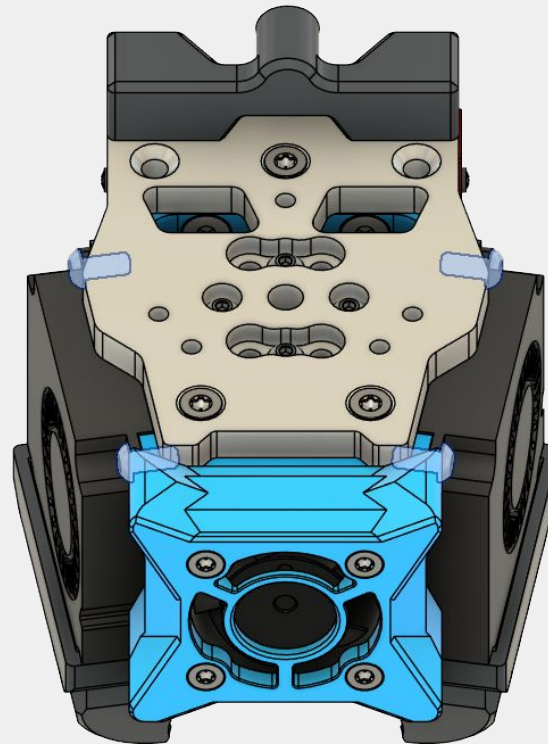


TIGHTEN TOP FAN-MOUNTING

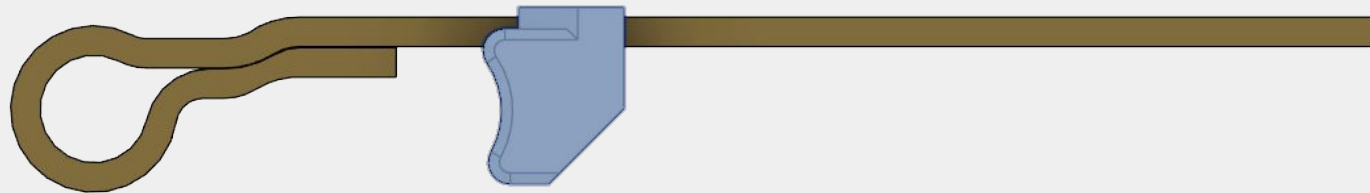
WWW.BEAVERTONMILESTONEHOBBY.COM

Finger tighten all four
M2.5x6 Button Head
Torx.

Note: don't over torque



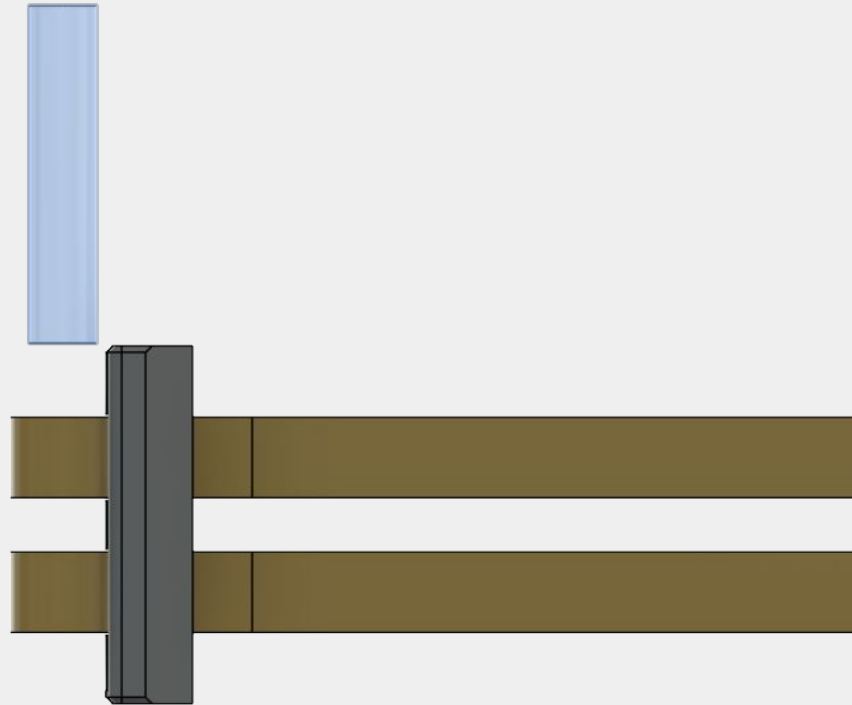
Insert the end of the belt into the belt tie then fold the belt the belt with the teeth together.



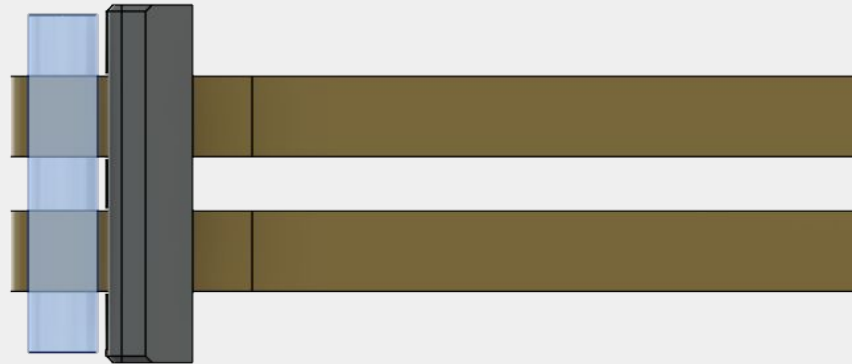
Slide the belt tie over
the folded over belt
loop.

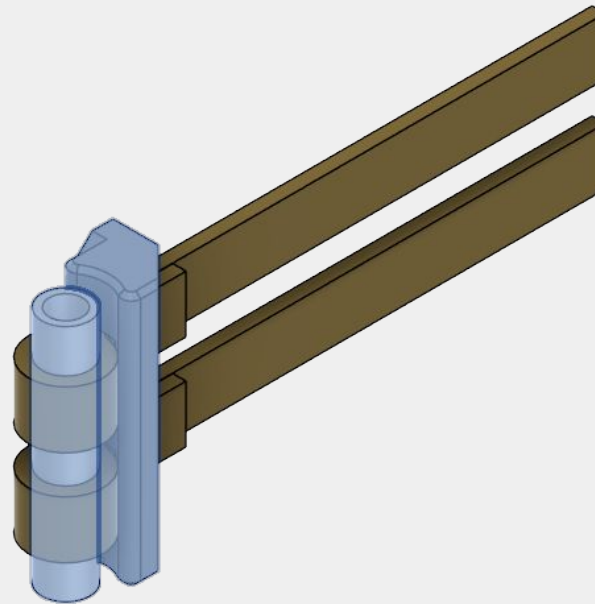


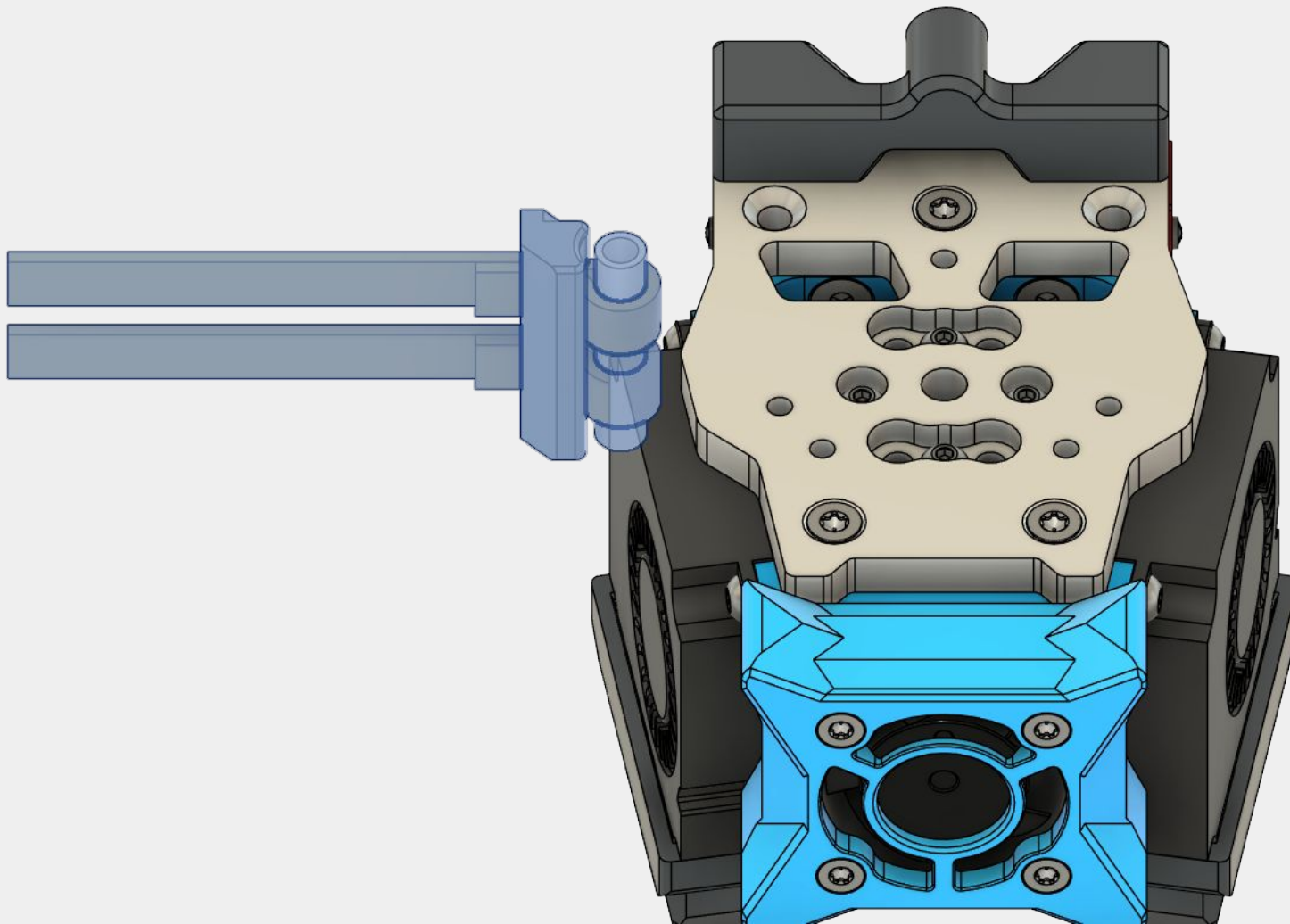
Slide the 25mm spacer
though the belt loop.



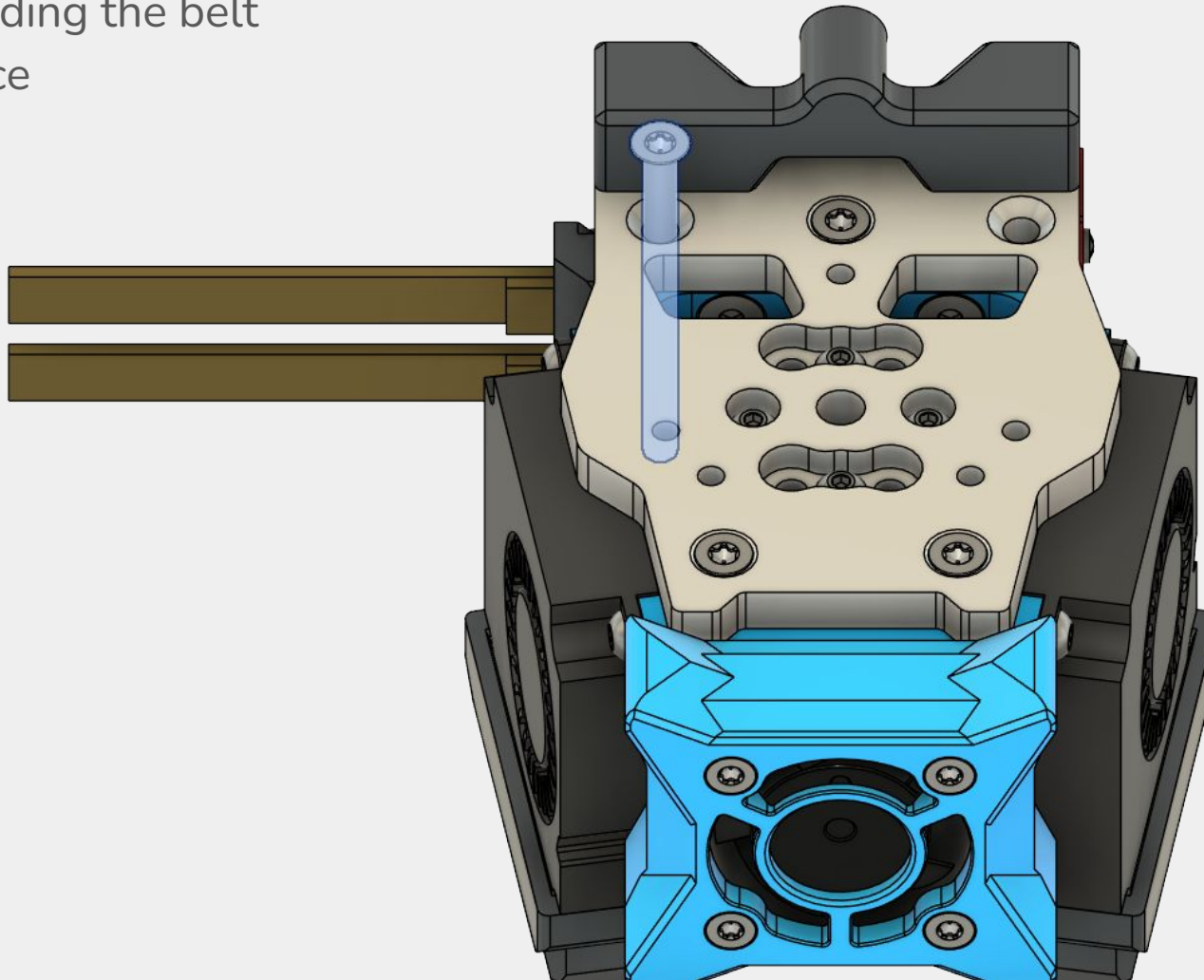
Center the spacer then
pull the belts tight over
the spacer.







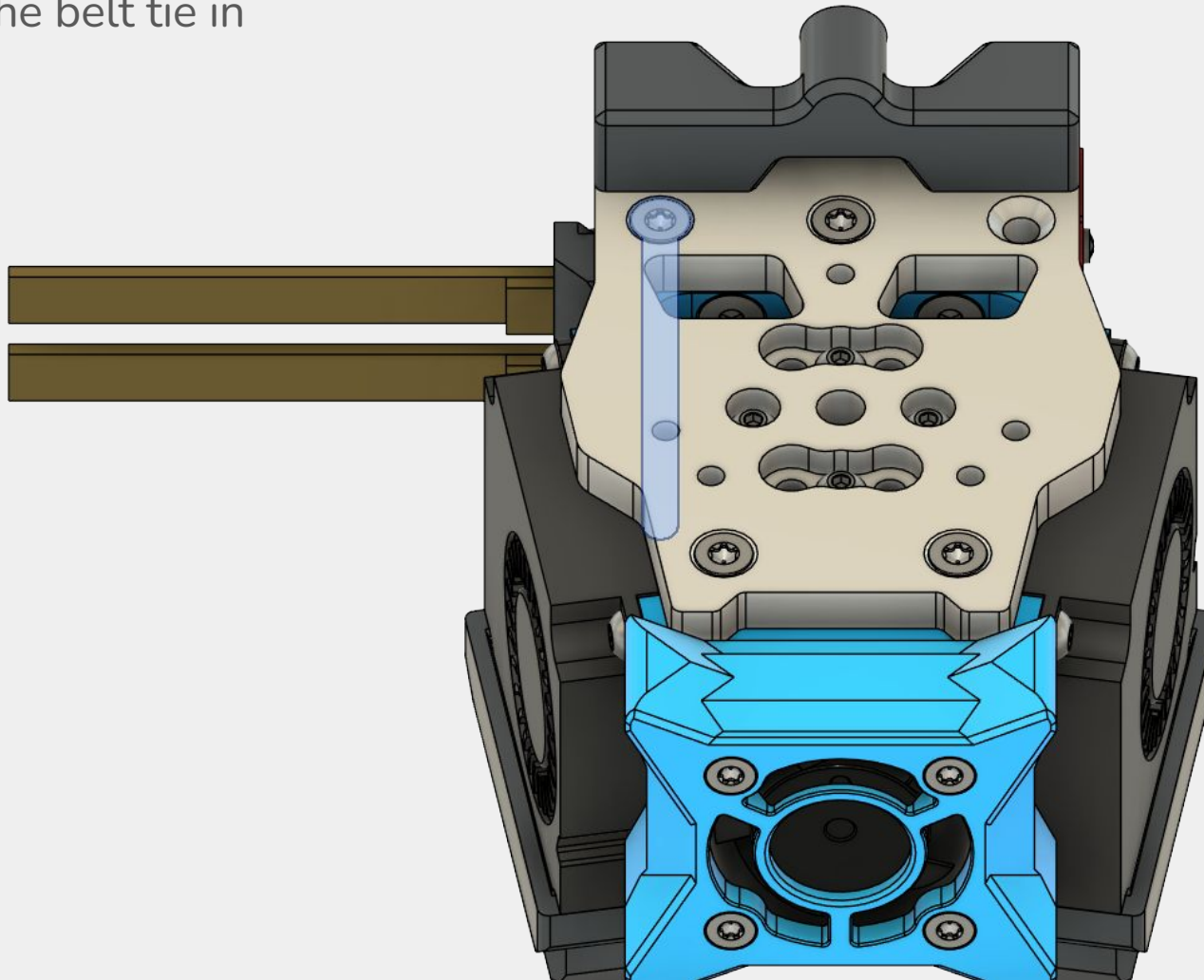
Insert a M3x40mm
countersunk Torx
while holding the belt
tie in place

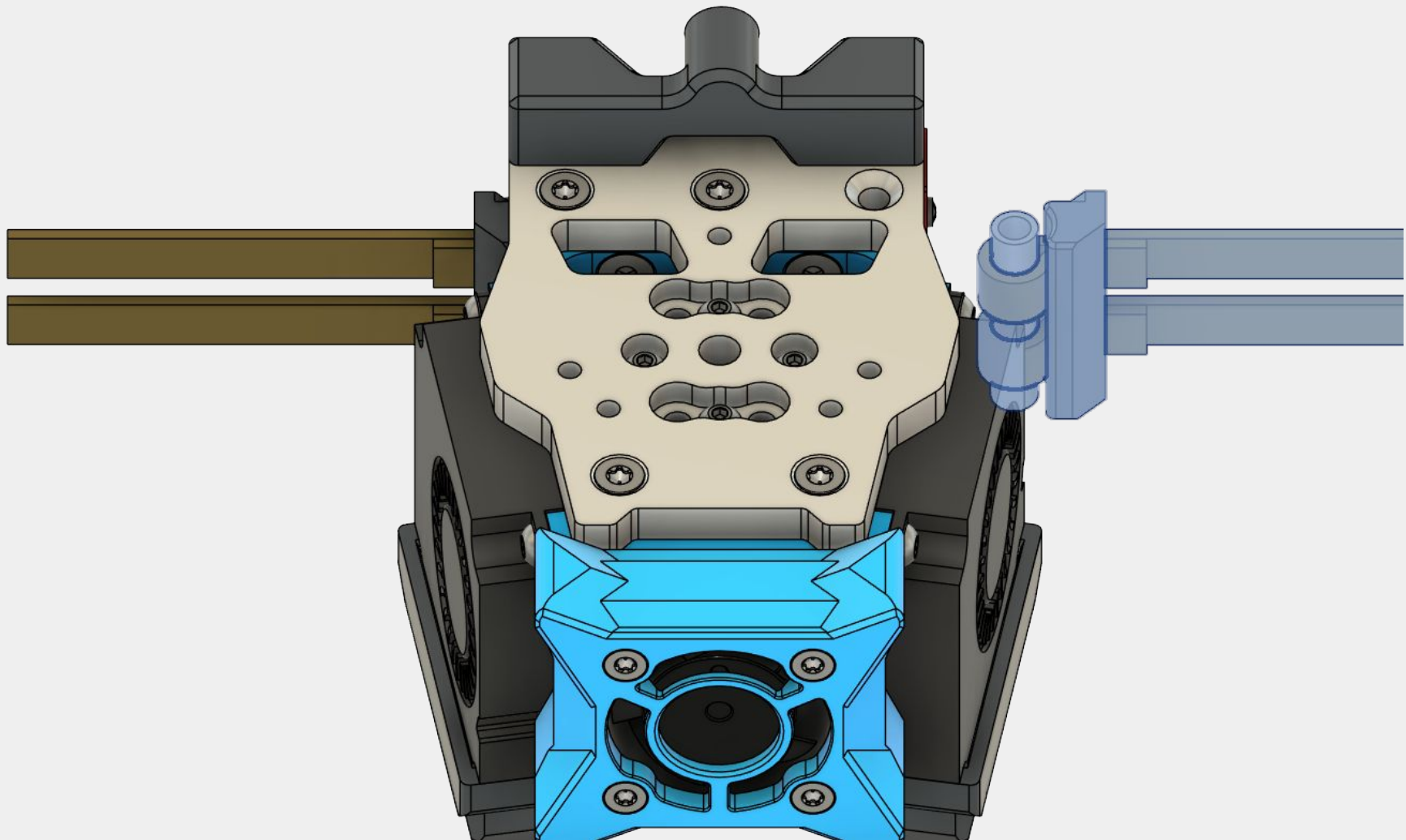


TIGHTEN TOP HARDWARE

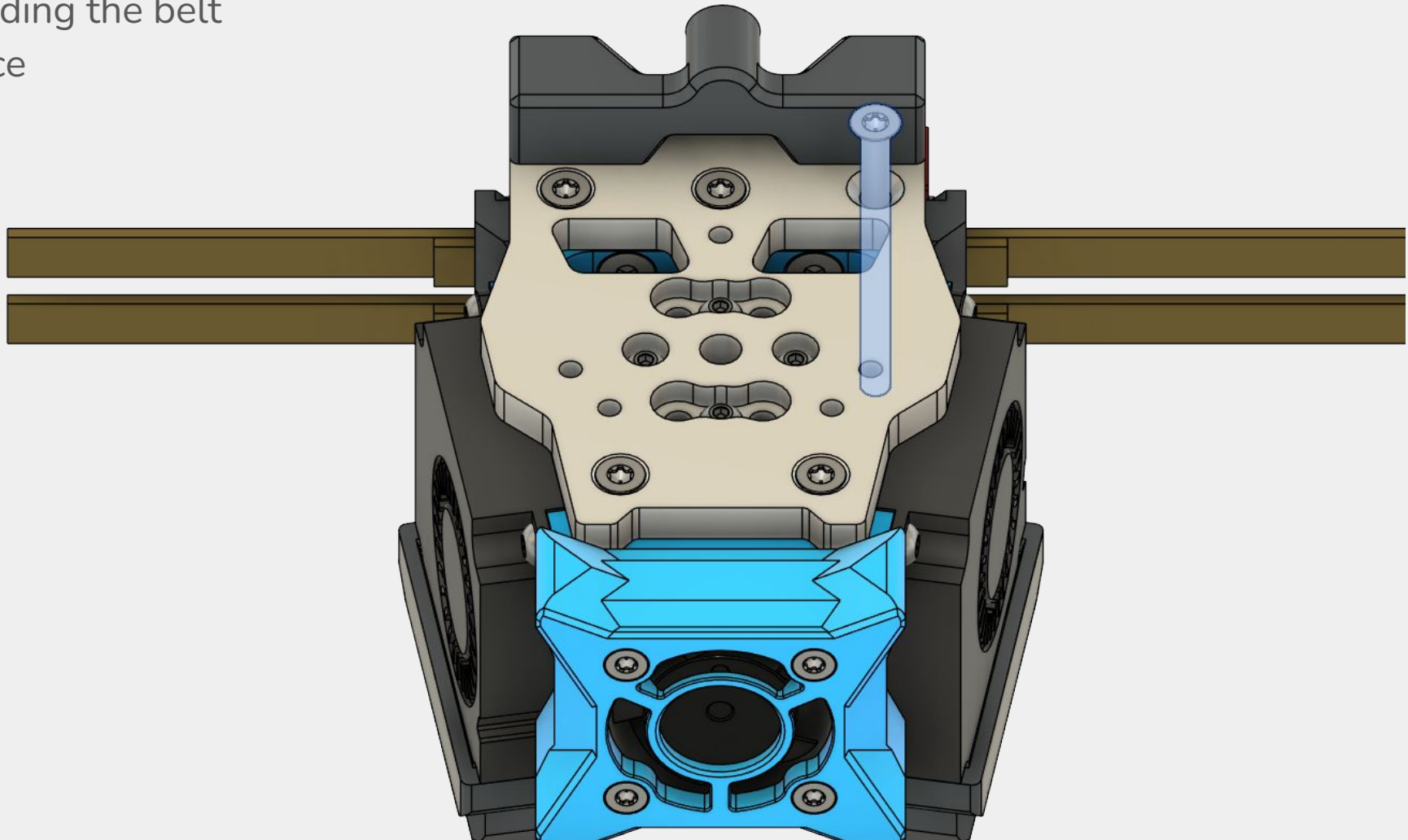
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Tighten the M3x40mm countersunk Torx while holding the belt tie in place





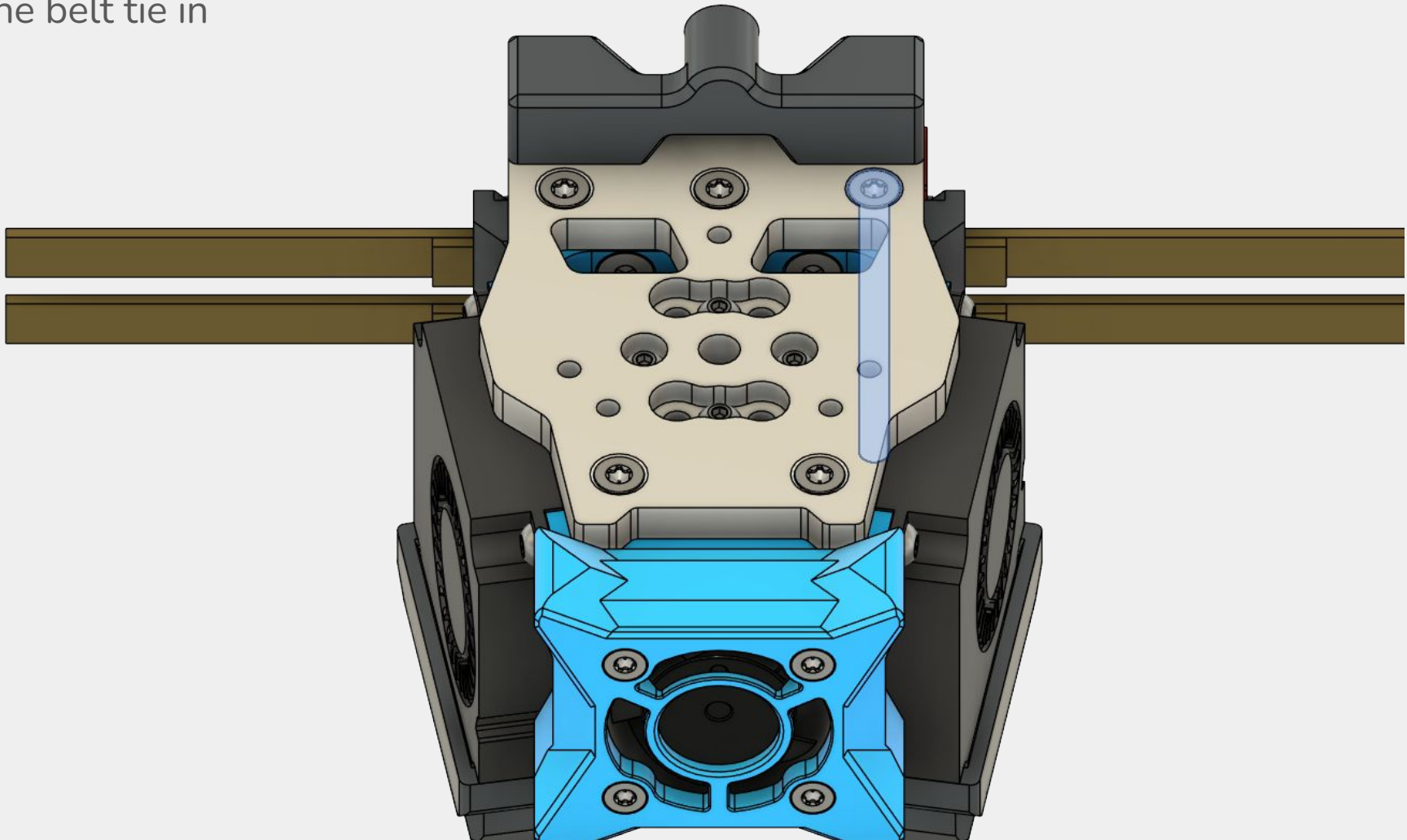
Insert a M3x40mm
countersunk Torx
while holding the belt
tie in place

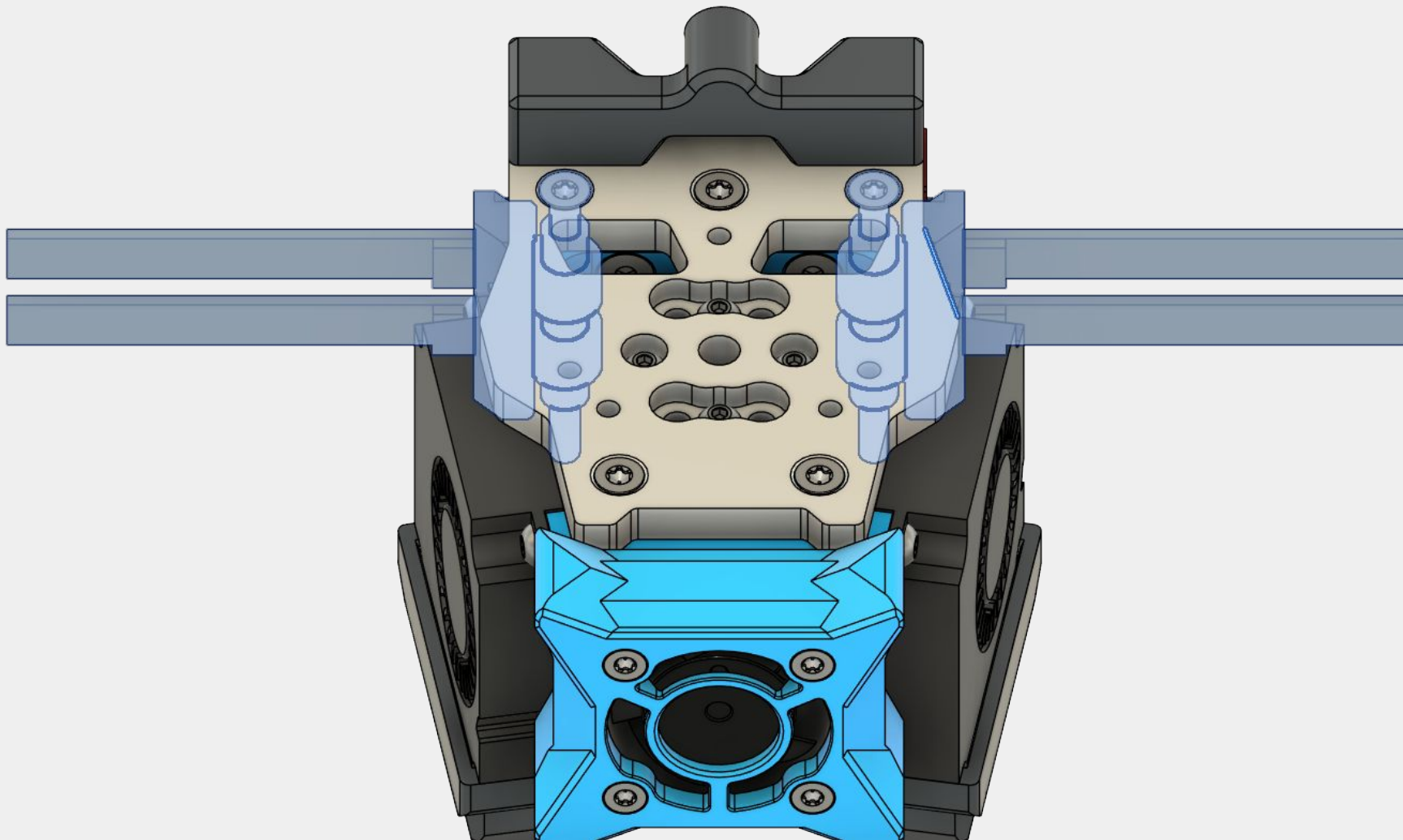


TIGHTEN TOP HARDWARE

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Tighten the M3x40mm countersunk Torx while holding the belt tie in place





REFERENCES

WWW.BEAVERTONMILESTONEHOBBY.COM

<https://github.com/Frix-x/klippain-shaketune>

<https://ellis3dp.com/Print-Tuning-Guide/>

Toolhead inspired by:

<https://github.com/Armchair-Heavy-Industries/Xol-Toolhead>

WEBSITE

WWW.BEAVERTONMILESTONEHOBBY.COM

GITHUB

<https://github.com>

DISCORD

<https://discord.gg/gZVXSz9fXj>